# TAU-ELICITATION: BENCHMARKING MULTI-TURN ENTITY EXTRACTION IN VOICE AGENTS

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## ABSTRACT

Voice agents often need to collect names, addresses, identifiers, dates, and times exactly, yet end-to-end benchmarks obscure where capture fails. We introduce τ-Elicitation, a 200-task voice benchmark spanning 10 entity types, controlled difficulty, caller realisms, and three environments. A matched text agent passes all tasks, but four voice configurations achieve robust exact success on only 0.14–0.41. Agents increase verification for hard and unfamiliar entities and sometimes for incorrect captures, but not for their weakest caller voice; only 24–37% of verified errors are repaired. A scaffold that prescribes spelling, read-back, correction, and confirmation raises robust $\mathrm{Pass}^3$ by 14–31 points, at a cost of 21–28 seconds per call. Realisms such as spelling variations and restarts do not detectably affect exact success; mispronunciation increases repair effort. These results identify strategy selection and successful recovery as the central bottlenecks in exact spoken entity collection.

***Index Terms*—** voice agents, spoken dialogue evaluation, multi-turn entity extraction, conversational repair, task-oriented dialogue

## 1. INTRODUCTION

Voice-agent benchmarks usually score complete tasks [1, 2, 3], yet authentication and key-entity transcription remain common failure points. Existing work does not isolate whether an interactive agent can recognize an uncertain capture, decide when to slow down, repair an error, and store the exact value. Nor does it measure how this capability changes with entity type, difficulty, repeated conditions, or caller effort. This matters because misunderstood named entities impede spoken dialogue, while confirming every attribute can itself frustrate callers [4].

τ-Elicitation tests this capability on short callbacks about missing database fields. Each core task elicits one missing value, isolating the atomic unit of spoken entity collection and establishing a minimum-complexity floor for broader calls; linked multi-entity tasks then test how errors compound. Its default prompt defines the *agent-directed* condition: the record must be exactly correct and letter-by-letter verification is suggested when unsure, but the agent decides when and how to verify. A matched *scaffolded* condition instead prescribes asking, spelling or reading back, confirming, correcting, and then submitting. Both share consent, never-guess, logging, and one-submission rules, isolating strategy selection from protocol execution. Deterministic database comparison scores both conditions without an LLM judge.

We contribute a reproducible 200-task benchmark spanning 10 entity types, controlled difficulty and caller realisms, and multi-entity calls. It measures capture, verification, repair, effort, and robust success across three environments; code and artifacts are available on GitHub.

**Table 1**. Coverage of related benchmarks. Task: executed tool task; Exact: exact entity score; $N$: controlled entity count; D: diagnostic or partial coverage.

| Work | Multi | Task | Exact | $N$ | Repair |
|---|---|---|---|---|---|
| Voice agents [1, 2, 3, 5] | ✓ | ✓ | D | – | D |
| Spoken dialogue [6, 7] | ✓ | – | ✓ | – | – |
| Speech NER [8, 9, 10] | – | – | ✓ | – | – |
| τ-Elicitation | ✓ | ✓ | ✓ | ✓ | ✓ |

## 2. RELATED WORK

Existing voice-agent benchmarks—τ-Voice, EVA-Bench, DuplexWorld, and Full-Duplex-Bench-v3—measure end-to-end task performance and interaction quality; several also exercise tool use [1, 2, 3, 5]. Their broad scope reveals failures such as authentication—identified as the dominant bottleneck in τ-Voice and reported separately in EVA-Bench—but does not isolate exact entity capture [1, 2]. τ-Elicitation extends the τ-bench, $\tau^2$-Bench, and τ-Voice lineage by making controlled voice entity collection the primary task [11, 12, 1].

Dialogue-state tracking spans MultiWOZ, SpokenWOZ, and cross-utterance sub-slot tracking [13, 6, 7]. ASR and spoken-entity work covers natural-speech NER, ASR error propagation, unseen entities in synthesized audio, and industrial missed-entity rates [8, 14, 9, 10]. *Error Detection and Recovery in Spoken Dialogue Systems* studies repair strategies [4]. None combines live tool execution, exact database writes, controlled entity type, difficulty and count, an agent's choice to verify. Table 1 summarizes this gap.

## 3. METHODS

### 3.1. Task and elicitation policy

Each task is a callback about a record with missing fields. The agent sees the field names, asks the caller for their values, logs its captures, and submits once; the caller reveals a value only after it is requested. The simulated caller is cooperative: it

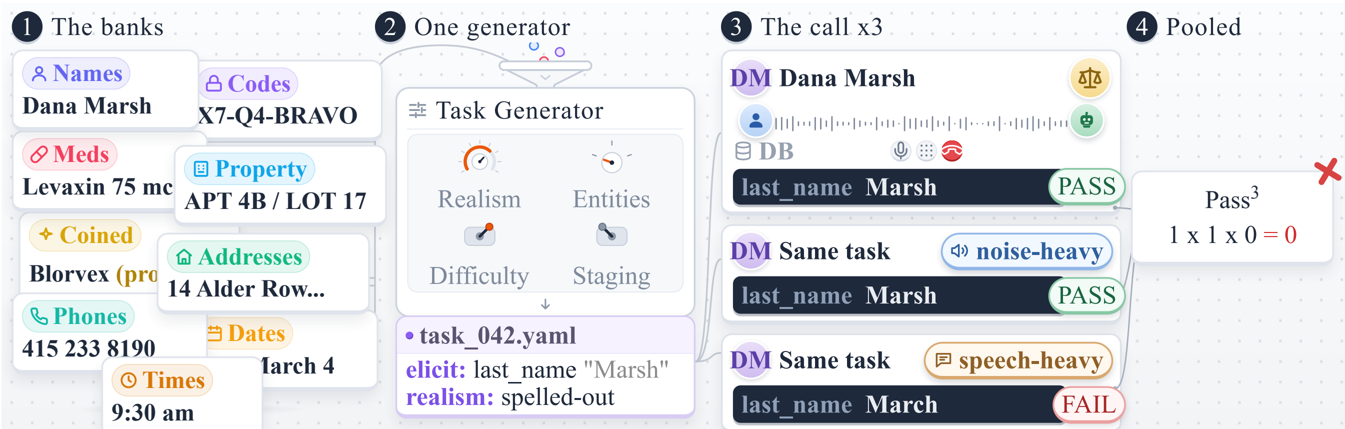


**Fig. 1**. τ-Elicitation selects values from entity banks, builds a controlled task, and runs it through the τ-Voice interaction loop. Pass$^3$ requires success in all three environment realizations. The scale marks fidelity-judged agent speech.

supplies requested values, spells accurately when asked, and confirms or corrects read-backs, but never volunteers information.

### 3.2. Values, difficulty, and realisms

Ten banks each contain 40 easy and 40 hard values; the benchmark samples 10 of each difficulty per bank. Difficulty is type-relative: for example, hard names are rarer, hard codes are longer or confusable, and hard emails contain more symbols. Table 2 reports bank construction and performance.

A seeded catalog adds six caller realisms by appending a scripted instruction to the caller prompt. For example, one spelling-style instruction tells the caller to say "zed" for the letter Z whenever spelling a value. Self-correction supplies a wrong value and immediately repairs it; spelling variation changes how letters or digits are grouped; falter-and-restart spelling introduces a mid-string correction; partial and wrong-field answers require a follow-up; and reviewed mispronunciations alter only the audio. Conditional behaviors occur only when the conversation creates an opportunity, so a spelling correction, for example, cannot fire until spelling begins. Each task stores its generator and catalog versions, seed, selected realism, and expected database state. The generator also emits linked two- and three-field tasks, both as flat calls and as a staged workflow that validates one field before revealing the next.

### 3.3. Scoring and behavioral measures

We compare each final database value with the exact gold string and report regular-realization success as Pass@1. We define robust Pass$^3$ as exact success on the same task across three prespecified environment realizations: *regular*; *noise-heavy*, with 10 dB background noise, bursts, frame drops, and muffling; and *speech-heavy*, with interruptions, backchannels, and restarts. Unlike a repeated-trial score, Pass$^3$ crosses these prespecified conditions; voices and realism assignments are also redrawn. It therefore asks whether the same task succeeds robustly, rather than averaging away a failure in one condition.

A 700-call audit placed the 90th-percentile duration at 1.89 minutes, so we use a conservative four-minute limit. Timed-out calls remain in the denominator and score zero.

We score both the first captured value and the final database value against the gold string. Silent caller tools log every spelling request and read-back; their sum is our measure of verification effort. We compare this effort across noise, entity difficulty and familiarity, and each system's best- and worst-performing caller voices.

## 4. EXPERIMENTS

**Systems.** We evaluate gpt-realtime-2 (minimal and xhigh reasoning), gemini-3.1-flash-live-preview (high), and grok-voice-think-fast-1.0 (provider default) [15, 16, 17]. The caller uses gpt-5.5 (xhigh, temperature zero) [15], with speech synthesized by ElevenLabs eleven_v3 [18].

**Matched runs.** Each configuration runs all 200 tasks under the regular, noise-heavy, and speech-heavy conditions. We repeat this grid with the scaffolded prompt, which prescribes spelling and read-back behavior. Within each condition, systems use the same tasks and random seed. This gives 600 agent-directed and 600 scaffolded calls per configuration, or 4,800 calls in total.

**Uncertainty and significance.** Confidence intervals come from 10,000 task-level bootstrap samples. When a task is sampled, its three linked conditions stay together. Pairwise system tests also align calls by task. We use a two-sided sign-permutation test with 100,000 permutations and the standard +1 correction. Holm correction is applied separately to the six system comparisons for Pass@1 and Pass$^3$.

**Human and judge validation.** Two raters independently review 90 failed calls across providers, labeling failure source and subtype. After discussion and six explicit adjudications, failure source is resolved for 86 of 90 calls and subtype for 66 of the 81 calls resolved as agent failures. For fidelity validation, gemini-3.1-pro-preview (temperature zero) scores a frozen stratified 60-utterance sample enriched for human flags [16].

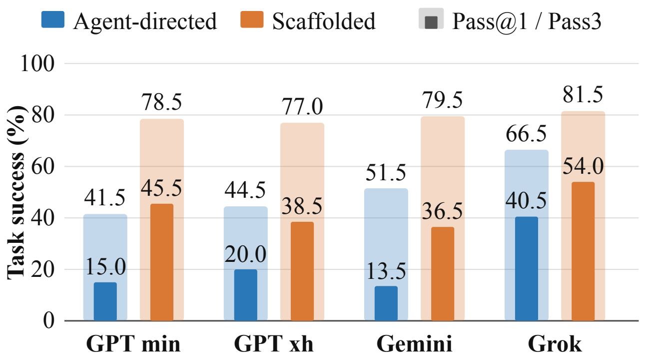


**Fig. 2**. Task success over the combined task set. Each pair shows agent-directed and scaffolded runs; light full bars show Pass@1, and dark inset bars show robust Pass$^3$.

## 5. RESULTS

### 5.1. Strategy selection and robustness

- *Scaffolding trades time for reliability.* It raises Pass@1 by 15–37 points and Pass$^3$ by 14–31 points (Figure 2), while adding 21–28 seconds per call. Gains are slightly larger for hard entities (27 versus 23 points).
- *A supplied protocol narrows provider gaps.* Grok remains best and significantly exceeds every other agent-directed system on both measures (Holm-adjusted $p \leq .0032$); no other pair differs. Systems execute a supplied protocol more reliably than they select one unaided.
- *Robustness exposes instability.* Across systems, Pass$^3$ spans 14–41% agent-directed and 37–54% scaffolded. Rankings can reverse: Gemini beats GPT xhigh on regular Pass@1 (51.5% versus 44.5%), yet trails on Pass$^3$ (13.5% versus 20.0%).
- *Diverse environments expose more than run-to-run variation.* For scaffolded GPT xhigh, three regular runs have similar Pass@1 rates (74.5%, 75.5%, and 77.0%), yet only 103 of 200 tasks pass every repeat (Pass$^3$ = 51.5%): 86 tasks change outcome at least once. Replacing two repeats with noise-heavy and speech-heavy conditions lowers the all-three count to 77 (38.5%). Environment diversity therefore reveals instability beyond stochastic reruns, and one-shot success overstates reliability.
- *More reasoning does not consistently improve robustness.* From minimal to xhigh, Pass$^3$ rises from 15.0% to 20.0% agent-directed but falls from 45.5% to 38.5% scaffolded.

### 5.2. Risk recognition and recovery

Initial captures are wrong for 69% of GPT xhigh fields, 65% of Gemini fields, and 42% of Grok fields. GPT and Gemini verify 66% and 64% of wrong captures versus 50% of correct ones; Grok verifies nearly everything (94%/96%). The systems likewise spend more effort on hard and unfamiliar entities, but not on their weakest caller voice, and only Grok reacts to noise (Figure 3). Risk recognition is real but incomplete.

Only 27%, 37%, and 24% of verified GPT xhigh, Gemini, and Grok errors end correct. Among 347 initially wrong captures, recovery is 10% without a repair step, 26% with read-back only, 24% with spelling only, and 35% with both. Because agents choose when to verify, these are associations, but the pattern locates the bottleneck after risk is noticed. Agents often double down on an error or hallucinate a replacement rather than complete the repair.

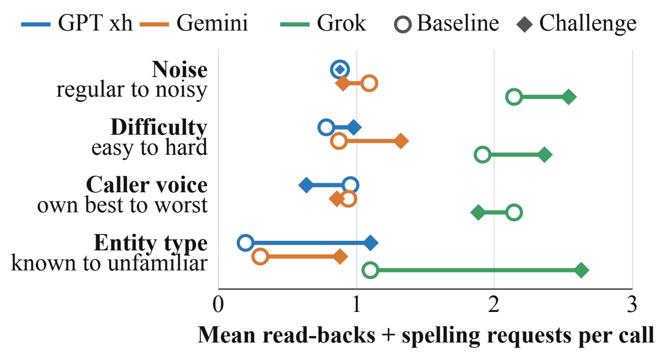


**Fig. 3**. Mean verification effort (read-backs plus spelling requests). All systems increase effort for hard and unfamiliar entities, but not for their own lowest-Pass@1 caller voice; only Grok increases effort under noise.

Caller voice is another blind spot. In agent-directed regular calls, the gap between each system's best- and worst-performing voices is 27 points for GPT xhigh, 28 for Gemini, and 15 for Grok. The ordering is not stable: Mildred leads GPT xhigh and Grok, whereas Wei leads Gemini. Only Gemini shows evidence that success differs across all five voices after correction ($p = .040$); no individual Gemini voice pair remains significant, so the result establishes heterogeneity without isolating one contrast. Across systems, Mildred significantly exceeds Priya, Mamadou, and Arjun, but not Wei. Yet no system increases verification for its own lowest-performing voice, suggesting that agents do not adapt their effort to voice-specific difficulty.

### 5.3. Stress tests and diagnostics

**Caller realisms mainly add repair cost.** No assigned realism detectably changes exact success in either arm (Figure 4). Overall Hájek estimates are $-3.6$ points (95% CI $[-10.4, 3.0]$) agent-directed and $-4.2$ ($[-8.9, 0.4]$) scaffolded. Assigned realisms may remain latent. In agent-directed runs, mispronunciation changes success by only $-0.5$ points but adds 24 points of spelling requests and 26 seconds. Realisms therefore add repair cost without explaining the large baseline failure rate.

**Entity structure determines difficulty.** Across the regular agent-directed cells, Pass@1 is 64.0% on easy entities and 44.3% on hard ones; scaffolding raises these rates to 87.3% and 71.3%. Pooled Pass$^3$ ranges from 1.7% for medications and 3.3% for coined names to 46.7% for phone numbers and 60.0% for dates (Table 2). The pattern is consistent with format-constrained versus lexical capture. Dates and phone

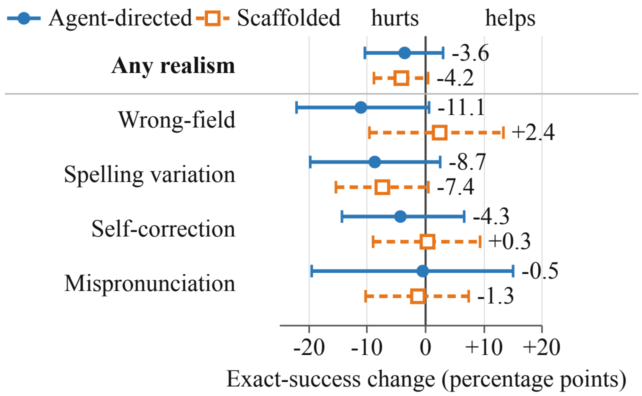


**Fig. 4**. Caller-realism effects in agent-directed (solid circles) and scaffolded (dashed squares) runs. Bars show task-clustered 95% Hájek intervals versus eligible clean assignments; none survives Holm correction.

**Table 2**. Value-bank construction and robust agent-directed results. Every bank has 80 values and contributes 20 tasks (10 easy, 10 hard). $\text{Pass}^3$ pools 60 system–task triples per bank.

| Bank | Construction / grounding | $\text{Pass}^3 \downarrow$ |
|---|---|---|
| Dates | Calendar and clock | .60 |
| Phones | Regulator-reserved ranges | .47 |
| Times | Calendar and clock | .43 |
| Person names | SSA + Census [19, 20] | .30 |
| Codes | ISO 3779 VINs; mod-97 IDs | .25 |
| Properties | Registry-screened synthetic | .18 |
| Addresses | Synthetic address grammar | .10 |
| Emails | DNS-nonresolving domains | .08 |
| Coined | Registry-screened synthetic | .03 |
| Medications | RxNorm-checked [21] | .02 |

numbers follow tightly scoped numeric schemas. Medications come from a closed RxNorm vocabulary and coined names from a synthetic registry, but the agent is shown neither list.

**Errors compound across fields.** At three fields, only 0.53 of calls pass even though individual fields remain 0.75 accurate (Figure 5): one wrong field fails the entire call. Intermediate validation and retry counter this accumulation, raising task accuracy from 0.64 to 0.82 across the matched multi-field tasks (Table 3). Because the intervention combines both mechanisms, we do not attribute the gain to either one alone.

**Human review.** Across 90 failed calls, agreement or adjudication assigns 81 failures to the agent and two to user-simulator logic; no call is jointly attributed solely to infrastructure. Among the 81 agent failures, agreed or adjudicated labels assign 42 as transcription, 16 as logical, six as VAD, and two as hallucination; the remaining 15 retain subtype disagreement. The speech-fidelity judge compares agent audio with its reference transcript, flagging mispronunciations, word additions, omissions, and clipping; severity $\geq 2$ marks a clear, potentially confusing mismatch. The frozen set yields .80 precision, .75 recall, and .77 F1. A matched text run passes all 200 tasks, implicating spoken capture and repair rather than task understanding.

**Table 3**. Effect of field validation and retry under the scaffolded prompt. Task Pass@1 requires every field to be correct; field Pass@1 scores individual values.

| Workflow | Task Pass@1 | Field Pass@1 |
|---|---|---|
| Batch submit; no field validation | $.64 \pm .06$ | $.78 \pm .03$ |
| Validate each field; retry errors | $.82 \pm .05$ | $.86 \pm .03$ |

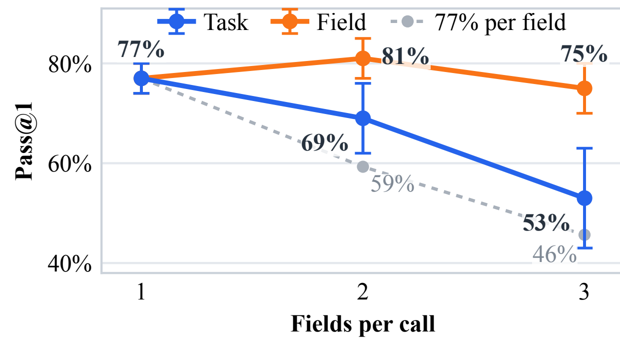


**Fig. 5**. Task success declines as fields accumulate, roughly following the light-gray reference expected if every field independently succeeds 77% of the time (77%, 59%, and 46%). Error bars show 95% confidence intervals.

**Table 4**. Share of scored agent utterances with a clear speech-fidelity defect by system (lower is better).

| | GPT minimal | GPT xhigh | Gemini high | Grok |
|---|---|---|---|---|
| Severity $\geq 2$ | 2.5% | 2.1% | 3.6% | 4.1% |

## 6. CONCLUSION

Voice agents still struggle to collect exact entities reliably. A strict elicitation protocol helps substantially, and our caller is deliberately highly cooperative: it spells accurately and corrects faulty read-backs. Even under these favorable conditions, scaffolded systems succeed across all three environments on only 37–54% of tasks.

Failures occur at both stages of recovery: agents do not consistently recognize risky captures, and when they do verify an error, only 24–37% are repaired.

**Limitations.** We study English, three providers, one simulator, and one run per main environment. We intentionally omit caller-side ASR to avoid confounding agent performance with a second recognizer. As a result, agent speech synthesis is not evaluated end to end: the speech-fidelity judge reports synthesis defects, but they do not yet affect task reward. The multi-field study is formative; broader provider, human-caller, and multilingual validation remain future work.

**Acknowledgment.** We thank Ben Shi, Ola Zytek, Vijay Iyengar, Ajeet Grewal, and Clay Bavor for feedback and support.

## 7. COMPLIANCE WITH ETHICAL STANDARDS

The benchmark uses no real customer records or recordings. Values are synthetic or public and non-sensitive; calls use simulated callers and provider voices. Reviewers consented and were compensated. Before release, artifacts will be screened for incidental personal data or secrets, and flagged items will be excluded.

Exact-value elicitation is dual use: it can improve service calls but can also facilitate collection of sensitive identifiers. Deployments should minimize collection, explain its purpose, protect stored values, and support human escalation. Our simulated English results are not evidence of safety, privacy, deployment readiness, or differences among people or demographic groups.

This work was supported by Sierra and Mercor. The authors are employees of Sierra and Mercor, respectively, and have no other relevant interests to disclose.

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